An Efficient IoT Model for On-Demand Particulate Matter Control System in Coal Mining Cities

Kumar Saurabh, Vikash, Lalita Mishra, Shirshu Varma
Department of Information Technology
Indian Institute of Information Technology, Allahabad
Prayagraj, Uttar Pradesh, India 211012
pwc2017001@iiita.ac.in, rsi2016503@iiita.ac.in, rsi2018502@iiita.ac.in, shirshu@iiita.ac.in

***Abstract*—Being the extension of IoT, Industrial IoT inherits (IIoT) the insecurities and vulnerabilities of the technology along with additional issues. Leveraging IIoT will have transformative effects on the industries with far greater efficiency, accuracy, scalability alongwith predictive maintenance However, this sensor-configured infrastructure of IIoT needs some extra efforts to customize the existing security solutions [1,2]. To address the entire security issues, a security framework is required which not only provides end to end security but also be trustable and resilient[3]. Though there are some effects but as of now no single security mechanism and framework is available to cater the entire IIoT ecosystem security issues. In view of recent advances the data-centric approach is being considered to serve as another alternative. Therefore instead of providing protection to different networks, communication technologies and protocols, a network layer based security framework is proposed which protects the data itself wherever it goes [4] and provides end to end security to the IIoT infrastructure.**

***Index Terms*—6LoWPAN, IoT, Particulate Matters, Coal Dust, Sensors**

I. Introduction

Industry 4.0 has generated a severe positive impact by incorporating IoT in every domain of industrial developments which consist of multiple Cyber-Physical Production Systems (CPPS), an integrated component of hardware, software, and communication devices and each layer is vulnerable to cyber-attacks. Existing security solutions are becoming insufficient as the resource constrained nature of heterogeneous devices and their networks are increasing with the popularity of IIoTs. To establish a secured automatic operation and avoid intermediate intrusions, a strong security system is required in IIoT whose security requirements are different from IoT security requirements [2]. . Lung carcinoma, asthma, ischemic heart disease and strokes, are the most detected diseases due to the exposure to particulate matters. Diameters of particles lesser that 10 μm are referred as PM_{10} and lesser than 2.5 μm are referred as $PM_{2.5}$ which are also called

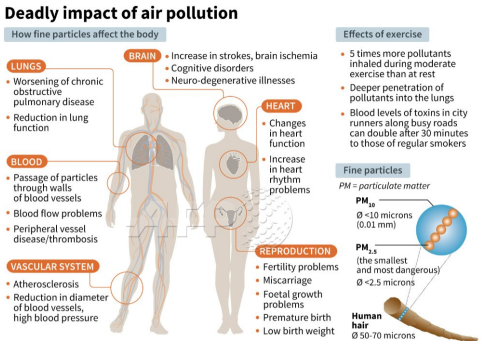


Fig. 1. PM Particles and their impacts, (Image Source: Agence France-Presse)

978-1-7281-6916-3/20/$31.00 ©2020 IEEE

fine particles, these particles are respirable and can be inhaled into the small boughs of air vessels in the lungs (Fig. 1) .Whereas particles with diameter lesser that 0.1 μm are referred as ultra-fine particles and they may enter into our blood [8], [9]. This paper proposes a framework to gather real-time particulate matter level data from sensors and to suppress the dust particles using IoT. The paper is organized into five sections; section 2 is about literature survey and hypothesis formation, section 3 is the proposed methodology, section 4 deals with results & discussions and section 5 discusses the conclusion and future work.

II. Literature survey & Hypothesis Formation

The present experiment is proposed for Dhanbad in Jharkhand (India). One of the primary control methods of dust control in mining cities is the misting of water to suppress the dust particles, These systems waste water a lot and their effectiveness varies application to application. In this manual water misting system, water tankers are used to spray water droplets on dust particles to get them down on the ground [10]. However, this method isn't much effective on respirable PM particles because the water pressure is generated by gravity and due to this, the produced water droplets size are greater than the respirable dust particles size. Therefore, new techniques are required to enhance the effectiveness of water misting on respirable dust particles as well as minimizing the wastage of water [11].

Hypothesis 1: Water Misting Technique helps to suppress coal, sand and other dust particles.

A lot of experiments [12] are performed and then the result is concluded that the pressurized water misting system significantly suppresses the respirable dust particles when compared to normal water misting systems. Although this pressurize water misting systems work efficiently on dust suppression but it is not possible to provide water misting all the time to maintain the PM level within the specified range, this will lead to start wasting of water.

Hypothesis 2: Pressurize water misting is the most effective misting system but it should be on demand.

It is observed that water misting process misspends the water and to avoid this, new technologies should be developed [12]. There should be some factors which help the system to start the misting process when required. Detection of vehicles and detection of hike in PM level may be the factors to decide the requirement of water misting. Number of vehicles definitely lead to increase in PM level but detection of a vehicle is not worthy if it is not going to be a cause for a significance change in the PM level.

Hypothesis 3: On Demand Water Misting (ODWM) should be required on the basis of detection of increment in PM level.

Sensors are energy and memory constrained devices. There are many communication protocols for these energy constrained sensors and 6LoWPAN is one of them which is an open standard protocol introduced and developed by IETF (Internet Engineering Task Force). Compared to other communication protocols available for energy constrained sensor nodes, 6LoWPAN has come out with the best performer in end to end delay, higher reliability (i.e. a lower % in packet loss) and lowest power consumption. 6LoWPAN enables each sensor node to connect with Internet via IPV6, it also provides 128 bit AES encryption for end to end security.

Hypothesis 4: 6LoWPAN is found the secure, energy saving and suitable communication protocol in the proposed model for the individual connection of ODWM SPOTs with Internet.

III. Proposed Methodology

Based on the first three hypothesis, we propose a framework for automatic suppression of the coal dust, sand dust and other respirable dust particles generated due to their transportation in coal cities. In the proposed framework, a concept of "ODWM" is proposed in which there is a set of various sensors (IR sensor, Rain sensor, GPS module, PM sensor, actuator (i.e, relay)) and Air-Water (i.e, pressurize water) misting mechanism are equipped to detect the need of water misting when

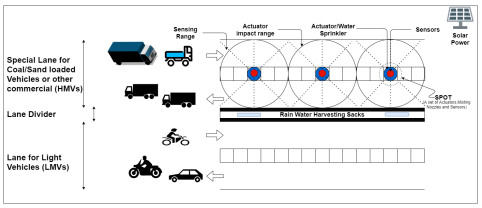


Fig. 2. Lane Design

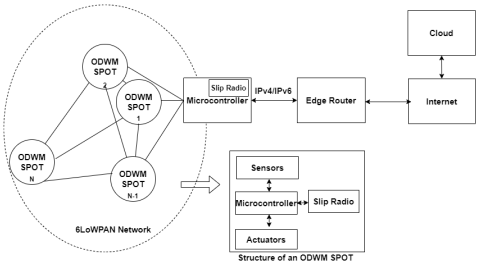


Fig. 3. Block Diagram

the particulate matter (specifically $PM_{2.5}$ and PM_{10}) level goes beyond the set threshold. As this framework is based on system demand, we named it "On demand water misting (ODWM)" system.
A separate lane design (Fig. 2) equipped with solar energy panels and rain water harvesting sacks is proposed for heavy motor vehicles (HMVs) and other cargo vehicles used for the transportation of coal and sand. These ODWM SPOTs are deployed on the lane divider at a fixed distance. When a vehicle enters in the sensing area of a SPOT and a significant rise in PM level is detected by the PM sensor then water misting automatically starts in case of no rain and automatically stops when PM level goes below the set threshold or in case of natural rain starts.

To reduce wastage of water and increase the dust suppression efficiency, there is a need to consider vehicle's speed as a deciding factor. The vehicle may enter in the sensing region with various speed modes. For example, if slow mode (e.g. speed of 0-10 km/hour) is detected then a set of SPOTs can be turn on sequentially, otherwise if all SPOTs are turned on then due to slow speed, vehicle can take a long time to reach to a long distanced SPOTs and lots of water will be misspend, similarly normal mode (e.g. speed of 10-20 km/hour) requires simultaneous mode of operation with a few numbers of SPOTs and fast mode (e.g, speed $>$ 20 km/hour) requires a large numbers of SPOTs should to be turned on simultaneously to increase the efficiency of dust suppression intelligently.

A. Block Diagram:

This block diagram (Fig.3) consists the structure of ODWM SPOT module and the block diagram of complete system. In the ODWM SPOT structure, the Raspberry Pi is equipped with sensors and actuator. This module also consists a 802.15.4 Radio to transform the RPi in a 6loWPAN transceiver. There may be N ODWM SPOTs present in environment, each ODWM SPOT can act like a sensing and actuation device with a capability to communicate with each other in a mesh topology. Now in the complete system diagram, we can see that these ODWM SPOTs are constituting a 6LoWPAN network and then connected to the sink micro-controller which is also connected with 802.15.4 slip radio and acts like a 6LBR. This sink is connected with edge router to get wired/wireless internet connectivity which will be required to send the data to cloud for the further data processing and analysis.

B. Architecture:

In this paper, the proposed IoT model is based on four layers [13]. These Layers (Fig.4) perform different roles in an IoT framework which are given below:

1) Perception Layer: All heterogeneous devices like sensors, actuators and the microcontroller board (RPi) are associated with this layer. This layer is responsible to deal with various types of heterogeneous data collected by sensors, temporary storage and preprocessing to convert the analog data to digital format.

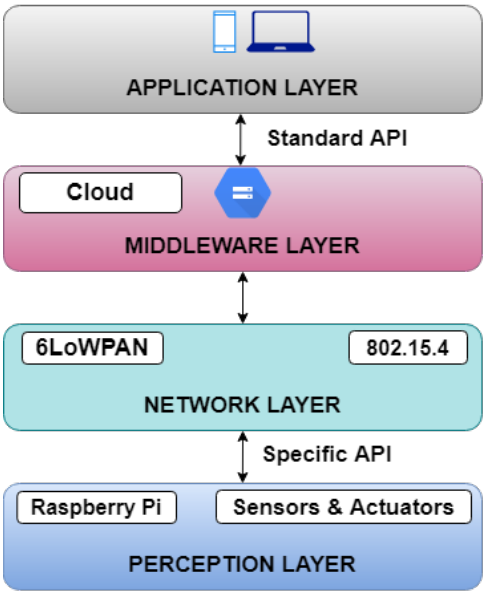


Fig. 4. Four Layer Architecture

2) Network Layer: This layer is responsible for transferring the data from perception layer to middleware layer and Application layer using various communication protocols like 6LoWPAN, Zigbee, 5G, 4G, Wi-Fi, Bluetooth, BLE, etc.

3) Middleware Layer: This is an abstraction layer which lies between network layer and application layer and facilitates the conversion of the gathered data from perception layer in a precise form as required by application layer. This layer also acts as a link between the application and smart things. In the proposed model, cloud is used to store the data collected by the sensors and sending it to the application layer.

4) Application Layer: This layer is used to provide the processed data as the result to the end user for further analysis through a website or an application or send this result to control actuators in an IoT application. In the proposed model, the application layer sends the processed data as a result to the website for monitoring purpose and to the micro-controller to turn on/off the actuator.

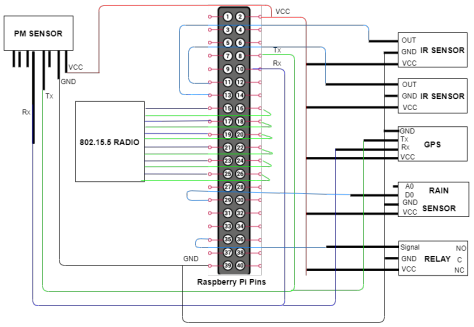


Fig. 5. circuit Diagram



Fig. 6. PM sensor, Rain sensor, IR sensor, relay, 802.15.4 radio, GPS module (From Left to Right)

C. Circuit Diagram & Components:

All sensors, actuator and 802.15.5 radio are connected to RPi pins as shown in the above diagram (Fig. 5). This circuit consists of a raspberry pi (model-3B) microcontroller board with two IR sensors for the detection and speed measurement of the vehicle. A GPS module is added to report the geographical location of the "ODWM SPOT". Rain sensor is used to detect rain. PM sensor is used to detect $PM_{2.5}$ and PM_{10} level. As an actuator, a relay is also added to raspberry pi to turn on/off the nozzle mechanism of "ODWM SPOT". The 802.15.5 radio is connected to pin 15 to 26 of Raspberry pi to convert the RPi in a 6LoWPAN border gateway router. The details of components (Fig. 6) used in this circuit are given below:

1) Microcontroller:: The microcontroller used in this circuit is Raspberry Pi model-3B (Fig.7). It comes with Broadcom BCM2387 chipset, 1.2 GHz Quad-Core ARM Cortex-A53 Processor, Wi-Fi, Bluetooth, 40 GPIO pins, etc. in a credit card size.

2) PM sensor:: The sensor used in this model is Plantower PMS5003 which finds the concentration

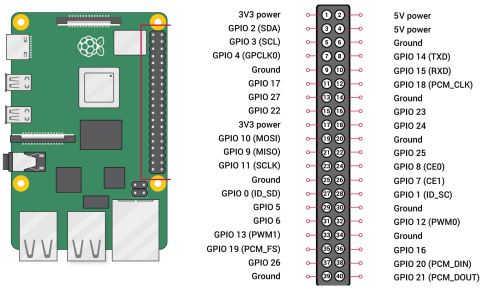


Fig. 7. Raspberry Pi model-3B

of particles and gives the result in digital format. An in depth study [14] has been done to find the best among various low cost particle sensors available like Alphasense OPC-R1, Novafitness SDS018, Honeywell HPMA115S0, Sensirion SPS030 and Plantower PMS5003. The Plantower PMS5003 is found the best performer after calibration when applied to sense PM particles of sizes $>0.3\ \mu m$.

3) Infrared (IR) Sensors:: This module detects the object. IR Sensor works in three stages. First stage contains infrared LED as infrared transmitter to produce an infrared ray which gets a reflection in case of an object is detected, Second stage contains Photo-diode as an Infrared Detector to receive the reflected infrared signal and third stage contains a LM358 Op-Amp as a comparator in control circuit to make the output high or Low.

4) Relay Module:: A two channel Relay Module is used in this circuit, which is SRD-05VDC-SL-C and it runs on 5 volts. It acts like a switch and controls a high current circuitry with a low current signal.

5) 802.15.4 Slip-Radio module:: To convert Raspberry Pi (RPi) in 6LBR (6LoWPAN border router), 802.15.4 Slip Radio module manufactured by Openlabs is used with it. This 802.15.4 Slip Radio module adds a support layer for 6LoWPAN over 802.15.4 to the RPi. A 6LoWPAN Border Router can perform translation of 6LoWPAN to IPv6 and the converse is also true. 6LBR also have the support for NAT64 (i.e, Network Address Translation from IPv6 to IPv4). A detailed procedure of convert a RPi into a 6LBR is given in [15]

6) GPS Module:: In this circuit, NEO-6M GPS Module is used. This module comes with a rechargeable button battery and a patch antenna for communication purpose.

D. Working Principle:

Scenario 1: When a vehicle is passing through the deployed area.

When the IR sensors detect a vehicle in its sensing range, then the microcontroller (RPi) sends instructions to the rain sensor to check the rain status. The rain sensor collects data and sends back to it. Now there may be two possible cases.

CASE I: If the rain is detected

If the rain is found then the microcontroller keeps all actuators in off state.

CASE II: If rain is not detected

If the rain is not detected by the Rain sensors then the microcontroller instructs the PM sensors to check the PM level. The PM sensor sends the data to it. If the PM level reading is less than the threshold set by the user then the microcontroller keeps all the actuators in 'off' state but when the reading is more than the threshold, then the microcontroller sends the instruction to the relay to turn on the water sprinkler mechanism. The microcontroller keeps on comparing the received data by the PM sensors and rain sensors to make the decision to keep the actuators in on state, until the PM level goes down to the set threshold or the natural rain starts. The microcontroller can choose three different modes of operation to turn on the actuators depending upon the speed of the vehicle detected in sensing area of the two IR sensors. These modes are namely Slow mode, Normal mode and Rapid mode. A user can set the parameters in the micro-controller program to turn on a number of actuators simultaneously or sequentially depending upon the vehicle's speed.

E. Algorithm:

Algorithm 1 PM level based Misting Process

Require: PM_value > PM_threshold
  while Rain is not detected **do**
    print (Alert", PM level is increasing)
    if vehicle_Speed > Speed_normal **then**
      switch on the relay in Rapid mode
    else
      if vehicle_Speed < Speed_normal **then**
        switch on the relay in Slow mode
      else
        switch on the relay in Normal mode
      end if
    end if
  end while

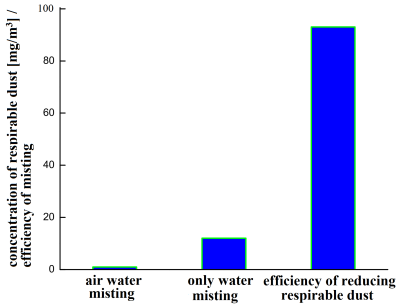


Fig. 8. Efficiency comparison graph of misting of air water and only water on respirable dust particles

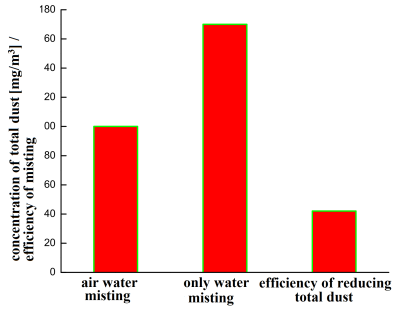


Fig. 9. Efficiency comparison graph of misting of air water and only water on total suspended particles

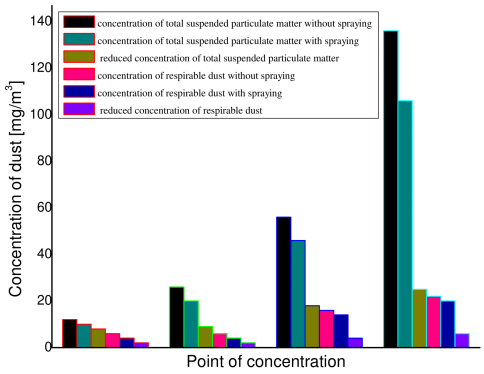


Fig. 10. Impact of misting on total dust and respirable dust at different points of concentration

IV. Results & Discussions

The proposed system uses air-water misting system which is very efficient to suppress the airborne dust when compared to manual water misting using tankers based on gravity water pressure technique [12]. In this paper, we have proposed an idea to automate this air-water misting process, which makes the complete system more advance, efficient, reliable, water conservative and on demand. This is why, the efficiency of the proposed IoT model will be definitely much higher than the air-water misting system alone.

The outcomes of the tests performed [12] to compare the efficiency of airborne total dust and respirable dust reduction using the air-water and only water misting system is shown in the figures (Fig. 8 & Fig. 9). To validate the versatility of the efficiency, multiple tests were performed on different points of dust concentrations but the recorded efficiencies were found quite similar (Fig. 10).

V. Conclusion and Future Works

A brief study on the issues of particulate matters, dust pollution and their consequences are presented in this paper. For any time, anywhere suppression of the pm particles along with coal dusts, an IoT architecture based on 6LoWPAN communication protocol is modeled which automates the air-water

misting process in case of a significant rise in pm concentration is detected. This framework is proposed to deploy in open world in order to solve the environmental safety concern, therefore huge security concerns are also associated with it. To provide an immunity to the system, we need to come up with effective solutions by performing rigorous attacks on the system and it's collected data. In our future work, We will implement this proposed application in the physical world to find it's various performance matrices and mitigation of possible security threats associated with it. The applicability of this research is limited for dust producing vehicles movement in separate lanes in coal cities. To address PM particles pollution issues in other cities, few modifications are still required.

REFERENCES

[1] S. Chowdhury, S. Dey, and K.R. Smith, "Ambient $PM_{2.5}$ exposure and expected premature mortality to 2100 in India under climate change scenarios," *Nature communications* vol. 9, no. 1, pp. 1-10, 2018.

[2] F. Karagulian, C. A. Belis, C. F. C. Dora, A. M. Prüss. Ustün, S. Bonjour, H. A. Rohani, M. Amann, "Ambient particulate matter (PM): A systematic review of local source contributions at global level," *Atmospheric Environment*,Vol. 120, no. 120, pp. 475-483, 2015.

[3] G. R. McGregor, D. Bamzelis, "Synoptic typing and its application to the investigation of weather air pollution relationships, Birmingham, United Kingdom," *Theoretical and Applied Climatology*, vol. 51, no. 51, pp. 223-236, 1995.

[4] S. Goswami, "Clean Coal Initiatives in India," *European Researcher*, Vol. 81, No. 8-2, pp. 1499-1513, 2014.

[5] GBD 2013 Risk Factors Collaborators, "Global regional and national comparative risk assessment of 76 behavioural, environmental, occupational and metabolic risks or clusters of risks in 188 countries 1990-2013: a systematic analysis for the GBD Background," *Lancet (London, England)*, vol. 386, no. 10010, pp. 2287, 2015.

[6] F.M.J. Bulot, S.J. Johnston, P.J. Basford, "Long-term field comparison of multiple low-cost particulate matter sensors in an outdoor urban environment," *Scientific reports*, vol. 9, no. 1, pp. 1-13,2019.

[7] M. Pedersen, L. G. Allemand, C Bernard, "Ambient air pollution and low birthweight: a European cohort study (ESCAPE)," *The lancet Respiratory medicine*, vol. 1, no. 9, pp. 695-704, 2013.

[8] K. H. kim, E. Kabir, S. Kabir, "A review on the human health impact of airborne particulate matter," *Environment International* vol. 74, pp. 136-143, 2015.

[9] N. Li, S. Georas, N. Alexis, "A work group report on ultrafine particles (American Academy of Allergy, Asthma & Immunology): Why ambient ultrafine and engineered nanoparticles should receive special attention for possible adverse health outcomes in human subjects," *Journal of Allergy and Clinical Immunology*, vol. 138, no. 2, pp. 386-396, 2016.

[10] Jharkhand State Polution Control Board, "Action Plan For improving Air Quality in Dhanbad," 2019. Accessed on: Nov. 18, 2020. [Online]. Available: https://bit.ly/3pGLvC7

[11] Y. S. XIE, G. X. FAN, J. W. DAI, X. B. SONG, "New Respirable Dust Suppression Systems for Coal Mines," *Journal of China University of Mining and Technology*, vol. 17, no. 3,2007, pp. 321-325, 2007.

[12] D. Prostanski, "Use of Air-and-Water Spraying Systems for Improving Dust Control in Mines," *Journal of Sustainable Mining*, vol. 12, no. 2, pp. 29-34, 2013.

[13] Vikash, L. Mishra, S. Varma, "Performance evaluation of real-time stream processing systems for Internet of Things applications," *Future Generation Computer Systems, Elsevier*, vol. 113,pp. 207-217, 2020.

[14] F. M. J. Bulot, H. S. Russell, M. Rezaei, M. S. Johnson, S. J. J. Ossont, A. K. R. Morris, P. J. Basford, N. H. C. Easton, G. L. Foster, M. Loxham, S. J. Cox, "Laboratory Comparison of Low-Cost Particulate Matter Sensors to Measure Transient Events of Pollution," *Sensors*, vol. 20, no. 8, pp. 2219, 2020.

[15] G. A. Naidu, J. Kumar, V. Garudachedu, P. R. Ramesh, "6LoWPAN border router implementation for IoT devices on RaspberryPi," *In Proceedings of 3rd International Conference on Internet of Things and Connected Technologies (ICIoTCT)*, pp. 26-27, 2018.